Sprint 3 Contribution Log - Kaila

Overview

In Sprint 3, I completed all of my database backend tasks: KA-9 (one login helper), KA-6 (notifications table, interview date column and migration), KA-7 (notification service, with the US-5 acceptance test drafts), KA-8 (demo data script) and KA-10 (requirements and README cleanup). Every task has its own branch, tests and pull request, the migration and the demo data are live on Neon, and the full test suite grew from 107 to 127 passing tests.

KA-9 - One Login Helper

Goal: Merge the two login helpers into one, so every protected route answers login problems the same way.

What I did:

Investigation: found that deps.py was used by the profile, resumes, applications and application status routers, and dependencies.py by the jobs and recruiter routers. No test or frontend file checked the error message, and the missing-header case behaved differently, so I used path C then A.

One helper: App/core/dependencies.py now uses HTTPBearer(auto_error=False) and answers 401 "Not logged in." for no header, a non-Bearer header, a bad or expired token, or a deleted account. require_recruiter is unchanged (403 "Recruiter access required").

Imports: pointed the four routers at App.core.dependencies and deleted App/core/deps.py.

Tests: tests/test_auth_helpers.py with 17 tests.

KA-6 - Notifications Table, Interview Date and Migration

Goal: Let the database store notifications and an interview date on each application, so the notification tasks (KA-7, KA-8, LA-7, LA-8, YO-7) can start.

What I did:

Notification model: SQLAlchemy model for notifications with user, type, message, application, read flag, creation time and event key (App/models/notification.py).

Data rules: deleting a user deletes their notifications, and deleting an application keeps the notification but clears the link. A unique rule on user and event key (uq_notification_user_event) stops the same reminder from being created twice.

Interview date: added interview_at to applications (App/models/application.py).

Model registration: imported the model so Alembic detects it (App/models/__init__.py).

Migration: wrote a migration that builds on 16d4c4e691bf and only adds the new table, its index and the new column (alembic/versions/1dae1f3df735_add_notifications_and_interview_at.py).

Tests: tests/test_notification_model.py with 5 tests.

KA-7 - Notification Service

Goal: Database functions for notifications, so the notification endpoints and the status change only need to call them.

What I did:

Service functions: create_notification, notify_new_application, notify_status_change, list_notifications, count_unread, mark_read and mark_all_read, with all the message wording in one file (App/crud/notifications.py).

No commit inside: the functions add and flush but never commit, so a notification is saved together with the change that caused it.

Ownership: mark_read returns nothing if the notification belongs to someone else.

Interview reminders (P2): generate_interview_reminders creates one reminder for each interview in the next 24 hours, with the interview time in the event key so a rescheduled interview reminds again. No background timer is needed.

Tests: tests/test_notifications_service.py with 15 tests.

Acceptance test drafts: wrote the three US-5 scenarios in Given / When / Then form: a status change notifies the job seeker, a new application notifies the recruiter, and an interview within 24 hours creates a reminder. Each one points to the service tests that cover it.

KA-8 - Demo Data Script

Goal: One command that fills the database with demo data, so the dashboard, the notifications and the recruiter screens have something to show in the demo.

What I did:

Demo script: a demo recruiter (Aurora Robotics) and a demo job seeker with hashed passwords, 12 jobs (5 posted by the demo recruiter), and 12 applications over the last 30 days covering all five statuses, one of them Interview with the interview tomorrow (scripts/seed_demo_data.py).

Notifications: created with my KA-7 functions: new applications for the recruiter, status changes, an interview invitation and an interview reminder for the job seeker.

Safety: shows the database host and only continues if you type YES, does nothing if the demo recruiter already exists, and commits once at the end. It never runs inside the tests.

README: added Step 7 explaining how to load the demo data.

KA-10 - Requirements and README Cleanup

Goal: requirements.txt has one line per package, and the README says only what really exists.

What I did:

requirements.txt: removed the duplicate pytest>=8.0.0 (kept pytest==9.1.1). httpx and httpx2 were both listed; nothing imports either directly, but the test client now uses httpx2 and warns when it gets httpx, so I kept httpx2 and removed httpx.

Features: the README listed things we don't have (AI feedback, CV matching, favourites). It now lists what exists, marks the Sprint 3 work as in progress, and puts the rest under Planned for Sprint 4.

API and tests: added an API section (all paths start with /api/v1, wrong input returns 400, docs at /docs) and updated Running Tests with the backend and frontend commands, the Windows command, a link to Documentation/TESTING.md and the two seed scripts.

Verification

All five tasks are tested and verified: the full suite went from 107 to 127 passing tests, the KA-6 migration is applied to Neon with alembic check reporting no differences, the demo data is loaded on Neon, and requirements.txt installs cleanly in a brand-new environment.

KA-9: grep shows nothing imports deps.py anymore. 124 tests passed (107 before plus 17 new), and the app started with uvicorn with no import errors.

KA-6: 112 tests passed. alembic heads showed one head. I told the team, previewed the SQL with alembic upgrade head --sql (only the notifications table, its index and interview_at), and applied it with alembic upgrade head. alembic current now shows 1dae1f3df735 and alembic check reports no new upgrade operations.

KA-7: 127 tests passed, including: the same event key twice gives one row, a rollback removes an unsaved notification, mark_read refuses another user's notification, and a reminder is created once for an interview 5 hours away and none for one 3 days away or a status other than Interview.

KA-8: typing "yes" in lowercase cancelled without changes. With YES, the first run added 2 accounts, 12 jobs, 12 applications, 9 notifications for the job seeker and 5 for the recruiter. The second run printed "Demo data already exists. Nothing was added." All 127 tests still pass.

KA-10: installed requirements.txt in a brand-new environment and ran the tests: 107 passed with httpx, but with a deprecation warning; with httpx2 instead, 107 passed with no warning.

Work Log

Nov 4, 2026 - Setup, KA-9, KA-6, KA-7, KA-8, KA-10: Set up the project on a second computer (venv, requirements, Git identity, .env). Merged the two login helpers (KA-9) and opened PR #___. Built the notifications table and interview_at column, applied the migration to Neon, and opened PR #___. Wrote and tested the notification service (KA-7), opened PR #124, posted the function names for Lamees and Youssef, and wrote the US-5 acceptance test drafts. Wrote the demo data script (KA-8), loaded the demo data on Neon, and opened PR #125. Cleaned requirements.txt and the README (KA-10), checked them in a clean environment, and opened PR #___. Hours: about 3.25 (setup 0.25, KA-9 0.75, KA-6 0.75, KA-7 0.5, KA-8 0.5, KA-10 0.25, acceptance test drafts 0.25).